

From WiFi Sensing to Quantum Sensing: Toward a Ubiquitous Sensing Theory

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Abstract

WiFi/4G/5G based wireless sensing has attracted a lot of attention from both academia and industry in the last decade. However, most of the work focused on developing effective techniques for a certain application, very few work attempted to explore the fundamental sensing theory and answer fundamental questions such as the sensing model, sensing limit, sensing boundary and sensing quality of WiFi/4G/5G signals. In this talk, I will first introduce the Fresnel zone model we proposed in UbiComp 2016 as a generic theoretic basis for contactless human sensing with WiFi/4G/5G signals, revealing the relationship among the received CSI signal, the distance between the two transceivers, the location and heading of the sensing target with respect to the transceivers, and the environment. Then I will present the Sensing Signal to Noise Ratio (SSNR) as a new metric to inform the sensing limit, sensing boundary and sensing signal quality of WiFi/4G/5G-based human sensing systems. In order to further increase SSNR and push the wireless sensing limit, we explore the Rydberg theory about how the Rydberg atom interacts with RF signals and develop the world first Rydberg quantum sensing system which senses tiny variations of RF signals caused by human activities, showing significant performance increase compared to WiFi or mmWave Radar based sensing systems.

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Short Bio:

Daqing Zhang is a Chair Professor at IP Paris and Peking University. His research interests include ubiquitous computing, mobile computing, big data analytics and AIoT. He has published more than 300 technical papers in leading conferences and journals, with a citation of over 33200 and H-index of 94. He developed the OWL-based context model and Fresnel Zone-based wireless sensing theory, which are widely used by pervasive computing, mobile computing, wireless networks and service computing communities. He was the winner of the Ten Years CoMoRea Impact Paper Award at IEEE PerCom 2013, and the Ten Years Most Influential Paper Award at IEEE UIC 2019 and FCS 2023, the Best Paper Award Runner-up at ACM MobiCom 2022, the Distinguished Paper Award of IMWUT (UbiComp 2021), etc.. He is now in the editorial board of ACM IMWUT, ACM TOSN and CCF TPCI. Daqing Zhang is a Fellow of IEEE and Member of the Academy of Europe.